

Binary Concentration Shift Keying Realization for Molecular Communication via Genetic Circuits

Rinrada Jadsadaphongphaibool
rinrada.jadsadaphongphaibool@kcl.ac.uk
King's College London
London, UK

Dadi Bi
dadi.bi@kcl.ac.uk
King's College London
London, UK

Yansha Deng
yansha.deng@kcl.ac.uk
King's College London
London, UK

ABSTRACT

The growth of synthetic biology has encouraged the molecular communication (MC) community to design biological-inspired systems. Synthetic biology provides tools to control and engineer living organisms to yield chemical signals relative to electronics signal. In this paper, we propose a binary concentration shift keying (BCSK) system for MC using genetic circuits. We characterize our proposed system by separating the system into three building blocks: molecule propagation to an absorbing wall, gate cells processing, and molecule propagation to the counting receiver. The analytical analysis of our proposed BCSK design can be derived from the combination of the building blocks' impulse responses. We validate the analytical derivations and the performance of the BCSK through BSim simulations.

CCS CONCEPTS

• **Hardware** → **Biology-related information processing**; • **General and reference** → Design; Validation.

KEYWORDS

Binary concentration shift keying, genetic circuits, modulation, molecular communication, synthetic biology

ACM Reference Format:

Rinrada Jadsadaphongphaibool, Dadi Bi, and Yansha Deng. 2022. Binary Concentration Shift Keying Realization for Molecular Communication via Genetic Circuits. In *The Ninth Annual ACM International Conference on Nanoscale Computing and Communication (NANOCOM '22)*, October 5–7, 2022, Barcelona, Spain. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3558583.3558857>

1 INTRODUCTION

Molecular communication (MC) is designed to facilitate the communications over chemical signals in biological and microscale environment. Over the past decade, the MC field has been centered around theoretical modeling using tools from communication engineering. This includes modeling propagation channels [1], improving transmission reliability through communication schemes [2], and evaluating the signaling pathway through information theory [3]. Those studies have demonstrated that MC can be applied

in various environments, e.g., human body and pipeline, which are limited by electromagnetic wave-based communication.

Despite the majority of research on theoretical modeling, signal processing realizations in the biological field have been overlooked. Synthetic biology aims to design and engineer biological components, novel devices, and systems as well as redesigning existing natural biological systems to achieve desired functions. Examples include genetic toggle switches [4], memory device [5], pulse generator [6], and digital logic gates [7, 8]. The breakthroughs in synthetic biology have inspired the research to perform computations using living organisms and encouraged the MC community to advance research in the biological domain.

The synthetic biology has been introduced to MC design recently. For example, signaling molecules C6-HSL were injected inside a microfluidic channel to be received by an engineered cell, which produced fluorescent light in the presence of C6-HSL molecules [9]. The system demonstrated that On-Off Keying is achievable for communication with a bacterial population by relying on molecular signaling in a microfluidic channel. In such case, microfluidic channels have allowed for dynamic changes in media composition, which is vital for maintaining the system, i.e., keeping the bacteria in an ideal condition and controlling the molecules' propagation direction. However, the study only focused on analyzing the communication capacity through simulations and fails to provide the analytical analysis of the MC design.

A biotransceiver incorporating genetically engineered bacteria tailored for MC was proposed to perform computational programming [10]. The design incorporates gene regulation to connect a building block to perform transceiver-like functionality. Although the design shows a promising application for transmitting signals with M-ary pulse amplitude modulation in MC, the implementation of the design is overlooked. Later on, the authors of [11] proposed a biotransceiver with parity-check encoding and decoding functions using genetic circuits. The study analyzed an end-to-end communication between the transmitter and receiver cells. Despite those mentioned above, the analysis used an approximation of the Hill function, which is far from the experiment and may hinder its practical application.

Different from the aforementioned studies, this paper aims to realize the binary concentration shift keying (BCSK) modulation and demodulation via genetic circuits in a microfluidic environment. Our design considers the implementation of the system by using a biological component that is available. The contributions of this paper are outlined as follows:

- (1) We propose a BCSK transceiver design based on existing and experimentally validated biological parts, including identity gates.

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NANOCOM '22, October 5–7, 2022, Barcelona, Spain

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ACM ISBN 978-1-4503-9867-1/22/10...\$15.00

<https://doi.org/10.1145/3558583.3558857>

- (2) We develop an analytical framework to characterize our proposed system. This is accomplished by dividing the system into building blocks, deriving the impulse response of each building block, and finally deriving the output expression of the end-to-end communication system. Our analytical results are validated by the BSim simulations, and provide a proof of concept for designing transceivers using genetic circuits.

The rest of this paper is organized as follows. Section 2 introduces the proposed BCSK system. In Section 3, we provide an analytical framework to characterize the proposed system. In Section 4, we describe the simulation framework and present the analytical and simulation results. Finally, we conclude this paper in Section 5.

2 SYSTEM MODEL

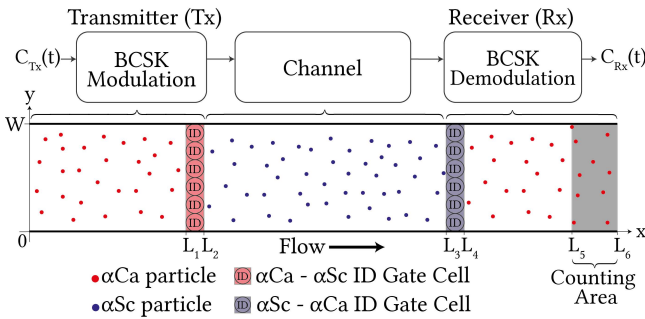


Figure 1: BCSK MC System

We present our proposed BCSK MC system in Fig. 1. It consists of two types of information molecules and two types of engineered yeast cells, which are capable of performing an identity logic (ID) function, i.e., producing a type of molecules if the concentration of the other molecule type is non-zero (bit-1). The α -factor mating pheromones from *Candida albicans* (α Ca) are added to the transmitter, denoted as $C_{Tx}(t)$. The molecules are carried by the fluid and propagate to the engineered α Ca- α Sc cells. The reception of α Ca molecules by α Ca- α Sc cells leads to the expression and release of the α -factor mating pheromone from *Saccharomyces cerevisiae* (α Sc) into the microfluidic propagation channel. In BCSK, bit-1 and bit-0 are represented by high concentration and low concentration, respectively. Due to the ID logic of the engineered cells, no molecules will be released for bit-0 transmission, thus achieving an on-off keying (OOK) function. Once the α Sc molecules arrive at the receiver, they can activate the engineered α Sc- α Ca cells to produce α Ca molecules, thus achieving the demodulation function. The α Ca molecules then propagate to a counting area, which generates the output of Rx, denoted by $C_{Rx}(t)$.

2.1 Microfluidic Channel Characteristic

For the 3D microfluidic channel, we assume that there is no diffusion along the vertical axis [6]. Thus, the molecule concentration can be described by the 2D advection-diffusion-reaction as [1, eq. (29)]

$$\frac{\partial C(x, y, t)}{\partial t} = D \nabla^2 C(x, y, t) - \nabla \cdot [u(x, y) C(x, y, t)] - k_{deg} C(x, y, t), \quad (1)$$

where D is the diffusion constant, ∇^2 is the Laplace operator, ∇ is the Nabla operator, $u(x, y)$ is the velocity, and k_{deg} is the degradation rate of the molecule.

The flow inside the microfluidic channel is assumed to be laminar, which is a valid assumption in microfluidic settings [12]. For the tractability of the analysis, we further assume a uniform flow with constant velocity u along the y -axis [13]. As a result, eq. (1) can be reduced to

$$\frac{\partial C(x, y, t)}{\partial t} = D \nabla^2 C(x, y, t) - u \frac{\partial C(x, y, t)}{\partial x} - k_{deg} C(x, y, t). \quad (2)$$

The molecules are released from $x = 0$ into the microfluidic channel at time $t = 0$. Thus, the initial condition can be defined as

$$C(x, y, t) = \delta(x) \Big|_{t=0}, \quad (3)$$

where $\delta(x)$ is the Dirac delta function. The boundaries of the microfluidic channel are assumed to be reflective, and we can define the boundary condition as [1, eq. (8)]

$$\frac{\partial C(x, y, t)}{\partial y} \Big|_{y=0, W} = 0. \quad (4)$$

2.2 ID Gate Cell

To implement the ID function, we borrow the regulated gene regulatory networks from [5] for the two types of engineered yeast cells. In particular, the regulated gene expression used in [5] is mainly consisted of three processes: *transcription factor binding*, *transcription*, and *translation*. First, α Ca or α Sc molecules act as a transcription factor and bind to the operator site. The operator site is a DNA sequence located before the transcription sequence, which marks the start of the transcription process.

As the binding process occurs on a much faster timescale than the subsequent transcription and translation processes, it can be treated in equilibrium and described by the Hill function as [14, eq. (7.7)]

$$\text{Fraction of bound operators} = \frac{[P]^n}{K^n + [P]^n}, \quad (5)$$

where $[P]$ is the concentration of the transcription factor, K is the disassociation constant, and n is the Hill coefficient. After the binding of transcription factor to operator site, the DNA is synthesized into mRNA through transcription, which will be further converted to protein p through translation.

As the timescale of transcription is faster than that of translation, the quasi-steady state is assumed, and the rate of protein production can be modeled as [14, eq. (7.10)]

$$\frac{dp(t)}{dt} = \beta \frac{[P]^n}{K^n + [\theta P]^n} - k_p p(t), \quad (6)$$

where β is the protein production rate, θ is a parameter corresponded to the data fitting, and k_p is the natural degradation rate of protein p during gene expression.

3 CIRCUIT ANALYSIS

In this section, we analyze the proposed BCSK system. To ease the analysis, we divide the end-to-end system into three building blocks as illustrated in Fig. 2:

- *Building Block 1*: molecule propagation inside a microfluidic channel and molecule reception by engineered cells;

- **Building Block 2:** the exchange of input molecules from extracellular to intracellular environment, the generation of output molecules through gene expression, and the release of output molecules from intracellular to extracellular environment;
- **Building Block 3:** molecule propagation inside a microfluidic channel for detection.

Specifically, we derive the impulse response of each building block in Section 3.1. Based on the derived impulse responses, we then analyze the transmitter, propagation channel, and receiver in Sections 3.2–3.4.

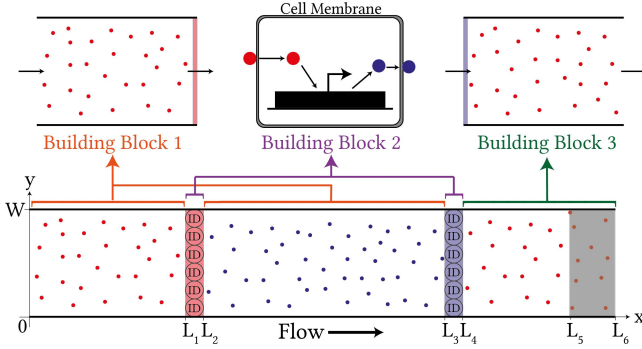


Figure 2: The building blocks of our proposed BCSK System.

3.1 Building Block Analysis

For all the derivations in Section 3.1, we denote $C_{Ii}(t)$, $h_i(t)$, and $C_{Oi}(t)$ as the input, impulse response, and output for building block i , respectively. Thereafter, the output of our proposed system can be written as the convolution of an input and a cascade of the impulse responses.

3.1.1 Building Block 1

In building block 1, molecules propagate inside the microfluidic channel and can be received by a column of well-aligned engineered cells, e.g., the engineered cells located at $L+R$, where L is the distance between the releasing point and the absorbing wall and R is the radius of the cell. The reception of αCa and αSc molecules relies on the receptors distributed on the surface of the cell membrane [15, 16]. Therefore, each engineered cell can be modeled as an active receiver and we assume the well-aligned cells create an absorbing wall along the y -axis. Moreover, for simplicity, we assume that the molecules will be absorbed at the left side of the cells (e.g., $x = L$) and the molecules generated by gene expression will be released at the right side of cells (e.g., $x = L + 2R$). Hence, the boundary condition can be defined as [16, eq. (5)]

$$D \frac{\partial C(x, y, t)}{\partial x} \Big|_{x=L} - k_a C(x, y, t) \Big|_{x=L} = 0, \quad (7)$$

where k_a is the absorption rate of one gate cell.

With the initial condition in (3) and the boundary conditions in (4) and (7), we derive the impulse response of (2) for building block 1 in the following theorem.

THEOREM 3.1. *The impulse response $h_1(x, y, t)$ of building block 1 in Fig. 2 is given by*

$$h_1(x, y, t) = 2e^{\frac{u(2x-ut)}{4D} - k_{deg}t} \sum_{n=1}^{\infty} \frac{(\lambda_n^2 + G_1^2) \cos(\lambda_n x)}{L(\lambda_n^2 + G_1^2) + G_1} e^{-\lambda_n^2 D t}, \quad (8)$$

where λ_n is the root of a transcendental equation $\lambda_n \tan(\lambda_n L) = G_1$, $\cos(\cdot)$ is a cosine function, and $G_1 = \frac{k_a}{D}$.

PROOF. See Appendix A. $\square$

THEOREM 3.2. *The concentration of absorbed molecules during time interval $[t, t + t_s]$ is derived as*

$$C_{O1}(t, t + t_s) = k_a \int_t^{t+t_s} C_{I1}(\tau) * h_1(L, \tau) d\tau, \quad (9)$$

where $C_{I1}(t)$ is input concentration of building block 1, $*$ is the convolution operator and t_s is the sampling time.

PROOF. The concentration of the molecules on the cell membrane can be expressed as

$$C(L, t) = \int_0^W C_{I1}(t) * h_1(L, y, t) dy. \quad (10)$$

To characterize the number of molecules absorbed on the absorbing wall, we define the change in the total number of particles due to the particle flux as [17, eq. (3.20)]

$$\frac{dN(t)}{dt} = \int_S D \nabla C(x, y, t) dS, \quad (11)$$

where S is the surface of the given region with differential element dS .

The change in the total number of particles in (11) on the absorbing wall is equivalent to the flux at the boundary specify in (7). By taking the integral of (11) at $x = L$ with respect to t , we can arrive at

$$N_{\text{react}}(t) = \int_0^t k_a C(L, \tau) d\tau. \quad (12)$$

Based on (12), the net change of absorbed molecules expected on the surface of the absorbing wall during the interval $[t, t + t_s]$ is defined as

$$\mathbb{E}[N_{\text{react}}(t, t + t_s)] = N_{\text{react}}(t + t_s) - N_{\text{react}}(t), \quad (13)$$

where $\mathbb{E}[\cdot]$ denotes the net change operation. Dividing (13) by W , we can arrive at (9). $\square$

3.1.2 Building Block 2

Building block 2 involves three processes: the *exchange* of input molecules from the extracellular to intracellular environment, the *generation* of output molecules through gene expression, and the *release* of the output molecules from intracellular to extracellular environment. Each process will be modeled by an ordinary differential equation (ODE) in the following.

When the external molecules are absorbed on the cell membrane, the molecules are transferred from the extracellular to the intracellular environment. The rate of the absorbed molecules is expressed as

$$\frac{dC_{Iin}(t)}{dt} = C_{I2}(t) - k_I C_{Iin}(t), \quad (14)$$

where $C_{Iin}(t)$ is the concentration of the absorbed molecules inside the cells, $C_{I2}(t)$ is the net concentration of the absorbed molecules, and k_I is the degradation rate.

LEMMA 3.3. *The impulse response of the transferring process in (14) of the molecules can be expressed as*

$$h_{21}(t) = e^{-k_I t}. \quad (15)$$

PROOF. Eq. (15) is the solution of (14) when $C_{I2}(t) = \delta(x)$. $\square$

As discussed in Section 2.2, the rate of protein production is expressed as

$$\frac{dC_{Oin}(t)}{dt} = \beta \frac{C_{Iin}(t)^n}{1 + [\theta C_{Iin}(t)]^n} - (k_O + k_d)C_{Oin}(t), \quad (16)$$

where $C_{Oin}(t)$ is the product molecules inside the cells, k_O is the degradation rate, and k_d is the rate of the product molecules released from the gate cell.

LEMMA 3.4. *The concentration of the product molecules inside the cells can be expressed as*

$$C_{Oin}(t) = \frac{\beta(1 - e^{-(k_O + k_d)t})}{\theta^n(k_O + k_d)} - \frac{\beta}{\theta^n(1 + [\theta C_{Iin}(t)]^n)} * e^{-(k_O + k_d)t}. \quad (17)$$

PROOF. Taking the Laplace Transform on (16), we obtain

$$\tilde{C}_{Oin}(s) = \frac{\beta}{s\theta^n(s + (k_O + k_d))} - \frac{\mathcal{L}\left\{\frac{\beta}{\theta^n(1 + [\theta C_{Iin}(t)]^n)}\right\}}{s + (k_O + k_d)}, \quad (18)$$

where $\mathcal{L}\{\cdot\}$ is the Laplace Transform operator. Taking the inverse Laplace Transform on (18), we can arrive at (17). $\square$

The rate of the release of molecules by engineered cells is expressed as

$$\frac{dC_{O2}(t)}{dt} = k_d C_{Oin}(t). \quad (19)$$

THEOREM 3.5. *The concentration of the product molecules emitted by the cells, $C_{O2}(t)$, is derived as*

$$C_{O2}(t) = \frac{\beta k_d \left[(k_O + k_d)t - 1 + e^{-(k_O + k_d)t} \right]}{\theta^n(k_O + k_d)^2} - \frac{\beta}{\theta^n \{1 + [\theta(C_{I2}(t) * e^{-k_I t})]^n\}} * \frac{k_d [1 - e^{-(k_O + k_d)t}]}{k_O + k_d}. \quad (20)$$

PROOF. With (18), we take the Laplace Transform of (19) and we can obtain the Laplace Transform of $C_{O2}(t)$ as

$$\tilde{C}_{O2}(s) = \frac{\beta k_d}{s^2 \theta^n [s + (k_O + k_d)]} - \frac{k_d \mathcal{L}\left\{\frac{\beta}{\theta^n \{1 + [\theta C_{Iin}(t)]^n\}}\right\}}{s [s + (k_O + k_d)]}. \quad (21)$$

Eq. (20) is obtained by applying the inverse Laplace Transform on (21). $\square$

Based on (20), the net change of product molecules' concentration released on the surface of the cells during the interval $[t, t + t_s]$ can be expressed as

$$\mathbb{E}[C_{O2}(t, t + t_s)] = C_{O2}(t + t_s) - C_{O2}(t). \quad (22)$$

3.1.3 Building Block 3

In building block 3, the molecules propagate inside a microfluidic channel without any obstacles. The main function of this building block is to detect the concentration of the output signals in a region. In practice, the concentration level can be visualized by fluorescent proteins or measured by electrochemical sensors, and we leave the analysis of these system for future work. As such, the molecule propagation can be modeled by the PDE (2) and the boundary condition (4).

COROLLARY 3.6. *The impulse response of building block 3 can be derived as*

$$h_3(x, y, t) = \frac{2}{\sqrt{4\pi Dt}} \exp \left[-\frac{(x - ut)^2}{4Dt} - k_{deg} t \right]. \quad (23)$$

PROOF. Eq. (2) can be transformed into a generic diffusion equation by transforming $C(x, y, t)$ into $F(x, y, t)e^{\frac{u(2x-ut)}{4D} - k_{deg} t}$ [18]. The solution to a generic diffusion equation with the conditions (3) and (4) is [17, eq. (3.31)]

$$C(x, y, t) = \frac{2}{\sqrt{4\pi Dt}} \exp \left(-\frac{x^2}{4Dt} \right). \quad (24)$$

Eq. (24) can be transformed back into solution for (2) as (23). $\square$

3.2 Transmitter Analysis

For the proposed BCSK transmitter, the net concentration of released molecules by the α Ca- α Sc cells at $x = L_2$ (see Fig. 1) during the interval $[t, t + t_s]$ can be expressed as

$$C_{Txout}(t, t + t_s) = C_{\alpha Sc}(t + t_s) - C_{\alpha Sc}(t), \quad (25)$$

where $C_{\alpha Sc}(t)$ is derived as (28) and shown in the top of next page. For (28), the $h_1(L_1, t)$ is

$$h_1(L_1, t) = 2e^{\frac{u(2L_1-ut)}{4D} - k_{\alpha Sc} t} \sum_{n=1}^{\infty} \frac{(\lambda_n^2 + G_1^2) \cos(\lambda_n L_1)}{L_1 (\lambda_n^2 + G_1^2) + G_1} e^{-\lambda_n^2 Dt}. \quad (26)$$

Based on (25), the net number of α Sc molecules released by the transmitter can be expressed as

$$N_{Tx}(t, t + t_s) = \int_0^W C_{Txout}(t, t + t_s) dy. \quad (27)$$

3.3 Channel Analysis

As shown in Fig. 1, the released α Sc molecules propagate inside a microfluidic and arrive at the proposed BCSK receiver. Since the output of the propagation channel with length $L_3 - L_2$ is the convolution of the transmitter output in (25) and the impulse response of the propagation channel in (8), we can write the the concentration of absorbed α Sc molecules at $x = L_3$ for the interval $[t, t + t_s]$ as

$$C_{Ch}(t, t + t_s) = k_a \int_t^{t+t_s} C_{Txout}(\tau) * h_1(L_3 - L_2, \tau) d\tau, \quad (31)$$

where $h_1(L_3 - L_2, t)$ is derived as (29) and shown in the top of next page.

$$C_{\alpha Sc}(t) = \frac{\beta_{\alpha Sc} k_d}{\theta_{\alpha Sc}^{n_{\alpha Sc}} (k_{\alpha Sc} + k_d)} \left\{ \frac{(k_{\alpha Sc} + k_d)t - 1 + e^{-(k_{\alpha Sc} + k_d)t}}{k_{\alpha Sc} + k_d} - \frac{1}{1 + [\theta_{\alpha Sc} k_a \int_t^{t+t_s} C_{Tx}(\tau) * h_1(L_1, \tau) d\tau * e^{-k_{\alpha Ca} t}]^{n_{\alpha Sc}}} * [1 - e^{-(k_{\alpha Sc} + k_d)t}] \right\} \quad (28)$$

$$h_1(L_3 - L_2, t) = 2 \exp \left\{ \frac{u[2(L_3 - L_2) - ut]}{4D} - k_{\alpha Sc} t \right\} \sum_{n=1}^{\infty} \frac{(\lambda_n^2 + G_1^2) \cos[\lambda_n(L_3 - L_2)]}{(L_3 - L_2)(\lambda_n^2 + G_1^2) + G_1} e^{-\lambda_n^2 D t} \quad (29)$$

$$C_{\alpha Ca}(t) = \frac{\beta_{\alpha Ca} k_d}{\theta_{\alpha Ca}^{n_{\alpha Ca}} (k_{\alpha Ca} + k_d)} \left\{ \frac{(k_{\alpha Ca} + k_d)t - 1 + e^{-(k_{\alpha Ca} + k_d)t}}{k_{\alpha Ca} + k_d} - \frac{1}{1 + [\theta_{\alpha Ca} (C_{Ch}(t, t + t_s) * e^{-k_{\alpha Sc} t})]^{n_{\alpha Ca}}} * [1 - e^{-(k_{\alpha Ca} + k_d)t}] \right\} \quad (30)$$

Table 1: The Parameters of the engineered ID gate cell [5].

Molecule	β (nM min ⁻¹)	θ (nM ⁻¹)	n	k (min ⁻¹)
αCa	0.0369	0.26	0.9	0.05
αSc	0.162	0.167	1.2	0.023

Table 2: The dimension of the microfluidic channel.

Dimension (μm)	W	L ₁	L ₂	L ₃	L ₄	L ₅	L ₆
	5	1	4	7	10	11	12

3.4 Receiver Analysis

For the proposed BCSK receiver, we derive the concentration of released αCa molecules by the αSc - αCa cells inside the counting area (see Fig. 1) with the input $C_{Ch}(t, t_s)$ as

$$C_{Rx}(t, t + t_s) = \mathbb{E}[C_{\alpha Ca}(t, t + t_s)] * h_3(x, y, t), \quad (32)$$

where $C_{\alpha Ca}(t)$ is derived as (30) and shown in the top of next page. For (30), the $h_3(x, y, t)$ has given in (23).

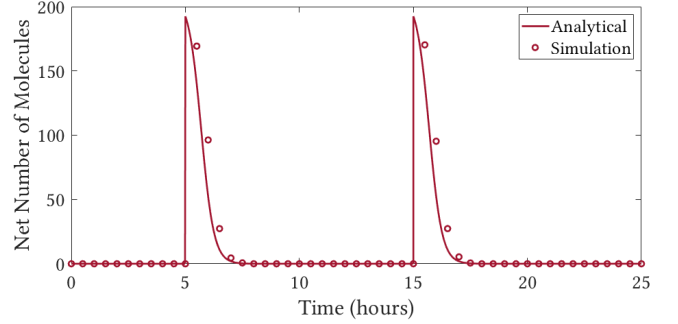
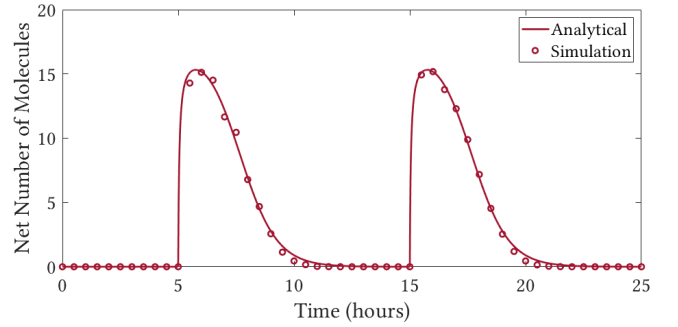
Based on (32), the number of molecules in the counting area is expressed as an integral over the entire area; the net number of αCa molecules in the counting receiver is expressed as

$$N_{Rx}(t, t + t_s) = \int_{L_5}^{L_6} \int_0^W \mathbb{E}[C_{\alpha Ca}(t, t + t_s)] * h_3(x, y, t) dy dx. \quad (33)$$

4 NUMERICAL RESULTS

In this section, we present the simulation results to validate our analysis and assess the performance of our proposed circuit. The simulations are implemented in BSim [19], which is a computational agent-based simulator for modeling cell dynamics in a typical environment. In BSim, the propagation of molecules is simulated using particle-based method [20, 16, 21], and each engineered cell is considered as an agent and simulated using an ODE solver. The simulation time interval is $t_s = 0.01$ s and the results are averaged over 2500 realizations. The parameters of the engineered ID gate cells and the dimension of the microfluidic channel are summarized in Table 1 and Table 2, respectively. The diffusion coefficient is set as $89 \mu m^2 s^{-1}$. The velocity and absorption rate are set as $20 \mu m s^{-1}$ and $10 \mu m s^{-1}$ respectively, so that the sum of the advection transport rate is less than 2 to ensure that transition probabilities are non zero [20]. The input of the proposed BCSK system is an impulse release of 250 molecules αCa molecules (i.e., 50nM) at $x = 0$ to transmit a bit-1; bit-1 is transmitted at time $t = 5$ hours and $t = 15$ hours.

Fig. 3(a) plots the analytical net number of molecules released by the transmitter at $x = L_2$ in (27) and the simulation results obtained from BSim. It clearly demonstrates a close match between

**(a) The net number of molecules released by the proposed BCSK transmitter.****(b) The net number of molecules released by the proposed BCSK receiver.****Figure 3: The numerical results of BCSK system.**

the analytical curves and the simulation points, which validates our theoretical analysis. We also observe that our proposed transmitter releases a certain number of molecules after the addition of 50nM of αCa molecules, and no molecules are released before $t = 5$ hours, thus achieving the OOK modulation.

Fig. 3(b) plots the analytical net number of molecules counted in the counting region in (33) and the simulation results obtained from BSim. We can see that the simulation results are close to the analytical results, which further validates our theoretical analysis. In addition, the receiver is also able to release some molecules representing bit-1 as a response to the transmitted bit-1 in Fig. 3(a). We also notice that the duration of the received signals is longer than the transmitter time interval. Hence, for this system, the transmission time interval should be at least 5 hours to avoid intersymbol interference.

5 CONCLUSION

In this paper, we proposed a BCSK transmitter and receiver design using genetic circuits. We derived the responses of the building blocks of our proposed system and combined the responses to derive the output expression of the end-to-end communication. We implemented our proposed design in BSim to demonstrate the BCSK modulation-demodulation functionality and validate our analytical framework. Our analytical framework provides a foundation for designing MC transceivers using genetic circuits. Our ongoing work is to extend the proposed genetic circuits to the realizations of higher-order CSK schemes and to analyze the communication aspect of the system.

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A PROOF OF THEOREM 3.1

We use the separation of variables method for THEOREM 3.1 by assuming $C(x, y, t) = X(x, t)Y(y)$. Eq. (2) can be separated into

$$\frac{\partial X(x, t)}{\partial t} = D \frac{\partial^2 X(x, t)}{\partial x^2} - u \frac{\partial X(x, t)}{\partial x} + (-k_{deg} + \gamma D)X(x, t), \quad (34)$$

$$\frac{d^2 Y(y)}{dy^2} - \gamma Y(y) = 0. \quad (35)$$

With separation variable, the initial condition (3) and boundary conditions (4) and (7) can be rewritten as

$$X(x, t = 0) = \delta(x), \quad (36)$$

$$Y(y) = 1, \quad (37)$$

$$\left. \frac{dY(y)}{dy} \right|_{y=0, W} = 0, \quad (38)$$

$$D \left. \frac{\partial X(x, t)}{\partial x} \right|_{x \rightarrow L} - k_a X(x, t) \Big|_{x \rightarrow L} = 0. \quad (39)$$

We first solve $Y(y)$ in (35) by considering three cases for γ , i.e., $\gamma > 0$, $\gamma = 0$, and $\gamma < 0$. According to the boundary condition at $y = W$, the solution for (35) does not exist when $\gamma > 0$ and $\gamma < 0$. Therefore, $Y(y) = 1$ with $\gamma = 0$.

Corresponding to the solution of $Y(y)$, γ term in (34) is removed. To solve $X(x, t)$, we first remove the advection and reaction terms. Specifically, the advection term is removed by setting $X(x, t) = F(x, t)e^{\frac{u(2x-ut)}{4D}}$, and the reaction term is removed by setting $X(x, t) = F(x, t)e^{-k_{deg}t}$ [18]. Then, eq. (34) becomes

$$\frac{\partial F(x, t)}{\partial t} = D \frac{\partial^2 F(x, t)}{\partial x^2}, \quad (40)$$

which is the general diffusion equation.

With the above transformations, the initial condition (36) and boundary condition (39) can be rewritten as

$$F(x, t = 0) = \delta(x)e^{-\frac{ux}{2D}}, \quad (41)$$

$$D \left. \frac{\partial F(x, t)}{\partial x} \right|_{x \rightarrow L} + k_{aT} F(x, t) \Big|_{x \rightarrow L} = 0, \quad (42)$$

where k_{aT} is the advection term of the transformed boundary condition.

The diffusion equation in (40) can also be solved by using the separation of variables method. With (42), the solution of (40) is

$$F(x, t) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n^2 D t} \cos(\lambda_n x), \quad (43)$$

where λ_n is the root of a transcendental equation $\lambda_n \tan(\lambda_n L) = \frac{k_{aT}}{D} = G_1$ and A_n is calculated, by applying the initial condition (41), as

$$A_n = \frac{2(\lambda_n^2 + G_1^2)}{L(\lambda_n^2 + G_1^2) + G_1}. \quad (44)$$

Finally, reverting the removed advection-reaction term to obtain $X(x, t)$ and by combining it with the solution of $Y(y)$, we derive the impulse response as (8).